S.C.O.U.T. Senior Design: Task Plan, Gantt Chart and WBS

Santa Clara Oceanic Utilities Transmitter · whole team · day-by-day detail Oct 26 – Dec 11, 2026, week overview through Jan 2027, milestones through May 2027 · built 2026-10-30 from `docs/planning/wbs-tasks.csv` (synced from Linear)

Team and ownership

- **John Ryan Myrdal (GENG)** — field and mechanical: buoy structure, deployment. 58 tasks.
- **Isabella Rodriguez (ECEN)** — hardware: electrical design, power, bench tests. 29 tasks.
- **David Chousal Cantu (CSEN)** — software: firmware, data pipeline, shore station. 13 tasks.
- Advisors: Jes Kuczenski (GENG), Shoba Krishnan (ECEN), Navid Shaghaghi (CSEN).

Milestones and external deadlines

- **Oct 08, 2026** — Component list delivered; unblocks housing dimensions *(Milestone)*
- **Oct 09, 2026** — Bench parts arrive (assumed Fri) *(Milestone)*
- **Oct 16, 2026** — Phase 1 exit: subsystem bring-up complete *(Milestone)*
- **Oct 18, 2026** — SCU: Undergraduate Programs Project Funding Request due; Gantt chart due *(SCU deadline)*
- **Oct 19, 2026** — Phase 2 starts: system integration *(Milestone)*
- **Oct 30, 2026** — Turbidity pod line complete (final pod soaked 1 week) *(Milestone)*
- **Nov 04, 2026** — First sensor working: temperature test demoed to the advisors *(Milestone)*
- **Nov 04, 2026** — Soak started: unit 1 in 35 ppt salt water (latest start to finish by 12/3) *(Milestone)*
- **Nov 25, 2026** — Phase 2 exit *(Milestone)*
- **Dec 04, 2026** — MVP functional prototype: floats, rights itself, reads one temperature and one turbidity value, packet at the shore station (target) *(Milestone)*
- **Dec 04, 2026** — FULL SYSTEM INTEGRATION (fall goal): sealed floating buoy reads temperature and turbidity, packet reaches the shore station and the dashboard *(Milestone)*
- **Dec 11, 2026** — MVP hard limit: before winter break *(SCU deadline)*
- **Jan 15, 2027** — Phase 3 exit: enclosure and waterproofing *(Milestone)*
- **Feb 01, 2027** — SCU: Senior Design Conference registration closes (winter) *(SCU deadline)*
- **Feb 26, 2027** — Phase 4 exit: field prototype deployed, two-week local test done *(Milestone)*
- **Mar 19, 2027** — Phase 5 exit: Hawaii deployment prep done, hardware shipped *(Milestone)*
- **Mar 22, 2027** — Hawaii live deployment starts (site and permits to be confirmed) *(Milestone)*
- **May 28, 2027** — Project end: final report and data from the deployment *(Milestone)*

How to read this

- **WBS:** 1 Project management · 2 Electrical · 3 Firmware & software · 4 Mechanical · 5 Deployment. Each task has a WBS number and, where one exists, its Linear issue.
- **Pages 2–5:** the day-by-day detail (Oct 26 – Dec 11, 2026), one block per work area, with dependency arrows. **Then:** the fall overview to Jan 15, and the full table with owner, dates, duration, dependencies, outside constraints and milestones for every task.
- **Dates:** solid bars end on the issue's Linear due date (a start is Linear's when the issue has started, otherwise proposed). Dashed bars are estimates; the table marks the source of every date. Parts arrival on Fri Oct 9 is an assumption to confirm.

John Ryan Myrdal (GENG) · Isabella Rodriguez (ECE) · David Chousal Cantu (CSEN) · dashed = estimated dates · purple dashed outline = dates John proposes for the owner to confirm (not yet in Linear) · red outline = blocked / waiting · → dependency (← lists the Linear issues a task waits for) · ◆ milestone · ◆ external deadline

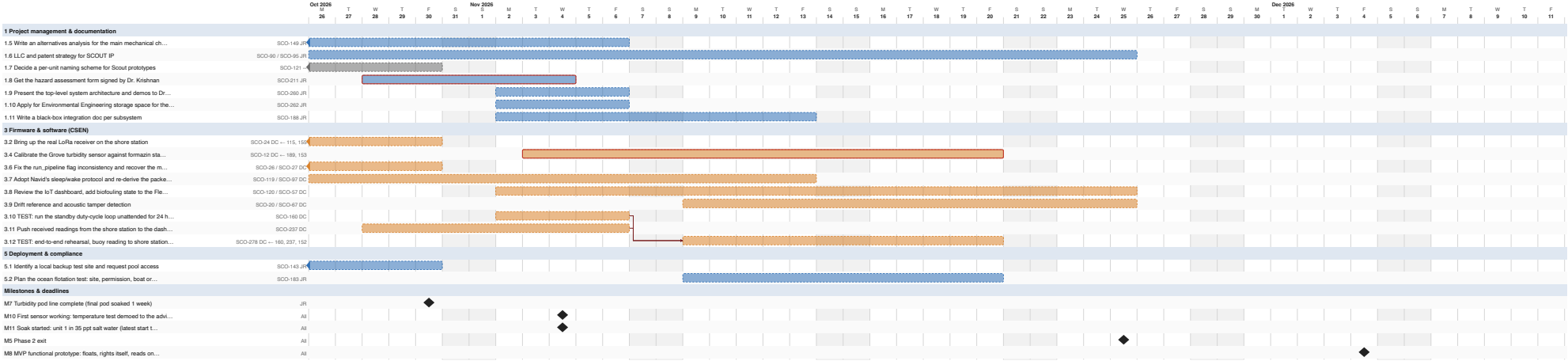
Outside constraints that drive the plan: SCU deadlines and course milestones; weekly team meeting (Thu); parts arrival and vendor lead times; shared printer access (industrial printer, Prusa XL, lab printer); advisor and reviewer availability; saltwater soak times of 24 hours to 1 week.

Detail: Oct 26 – Dec 11, 2026 · WBS 1 / 3 / 5 / Milestones & deadlines

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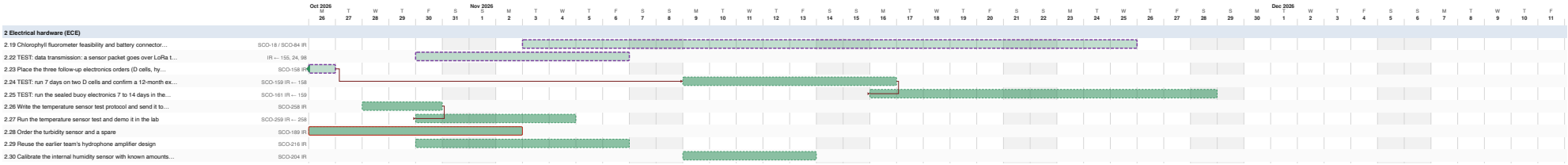


Detail: Oct 26 – Dec 11, 2026 · WBS 2

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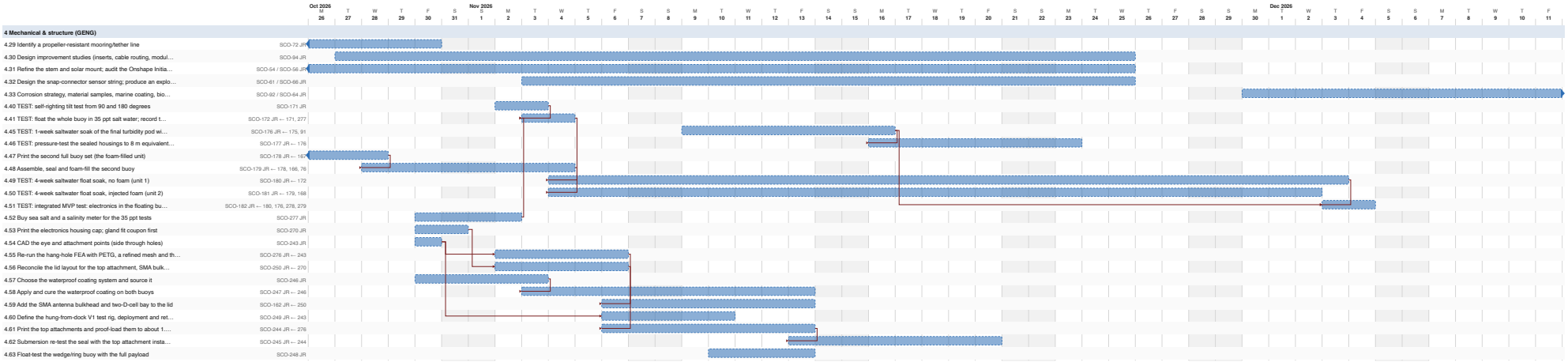


Detail: Oct 26 – Dec 11, 2026 · WBS 4

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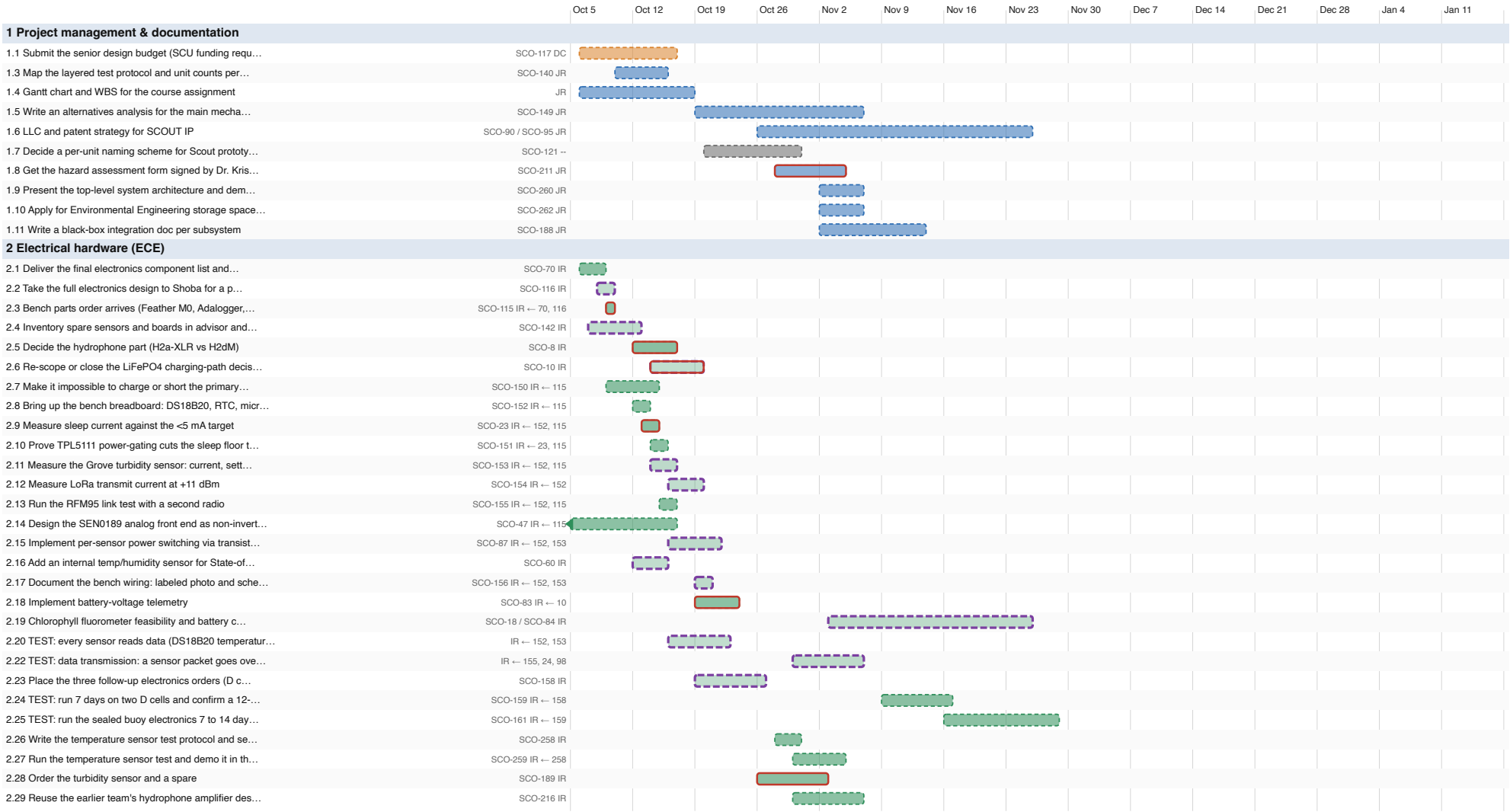


Fall overview: Oct 2026 – Jan 2027 (week columns)

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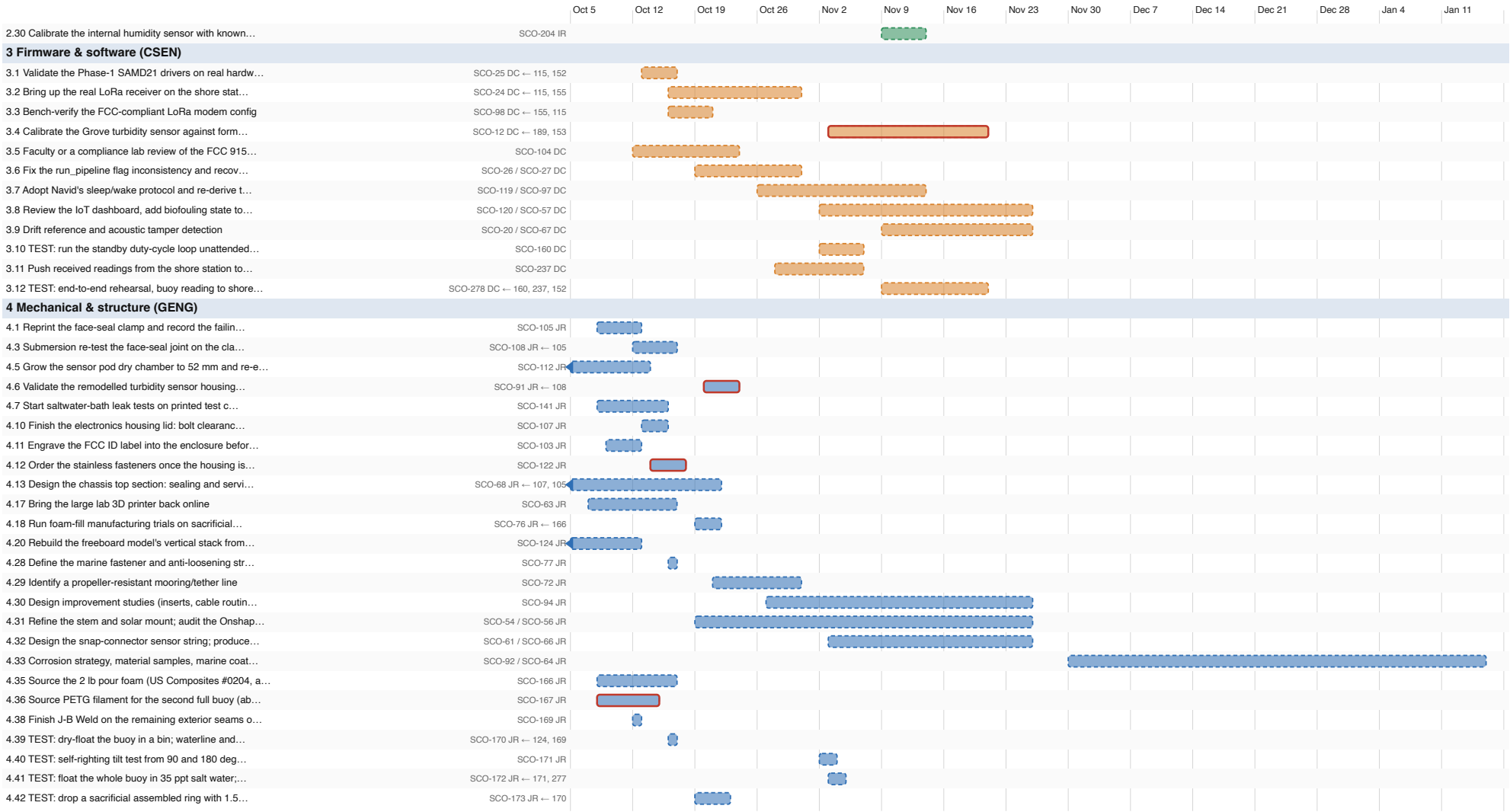


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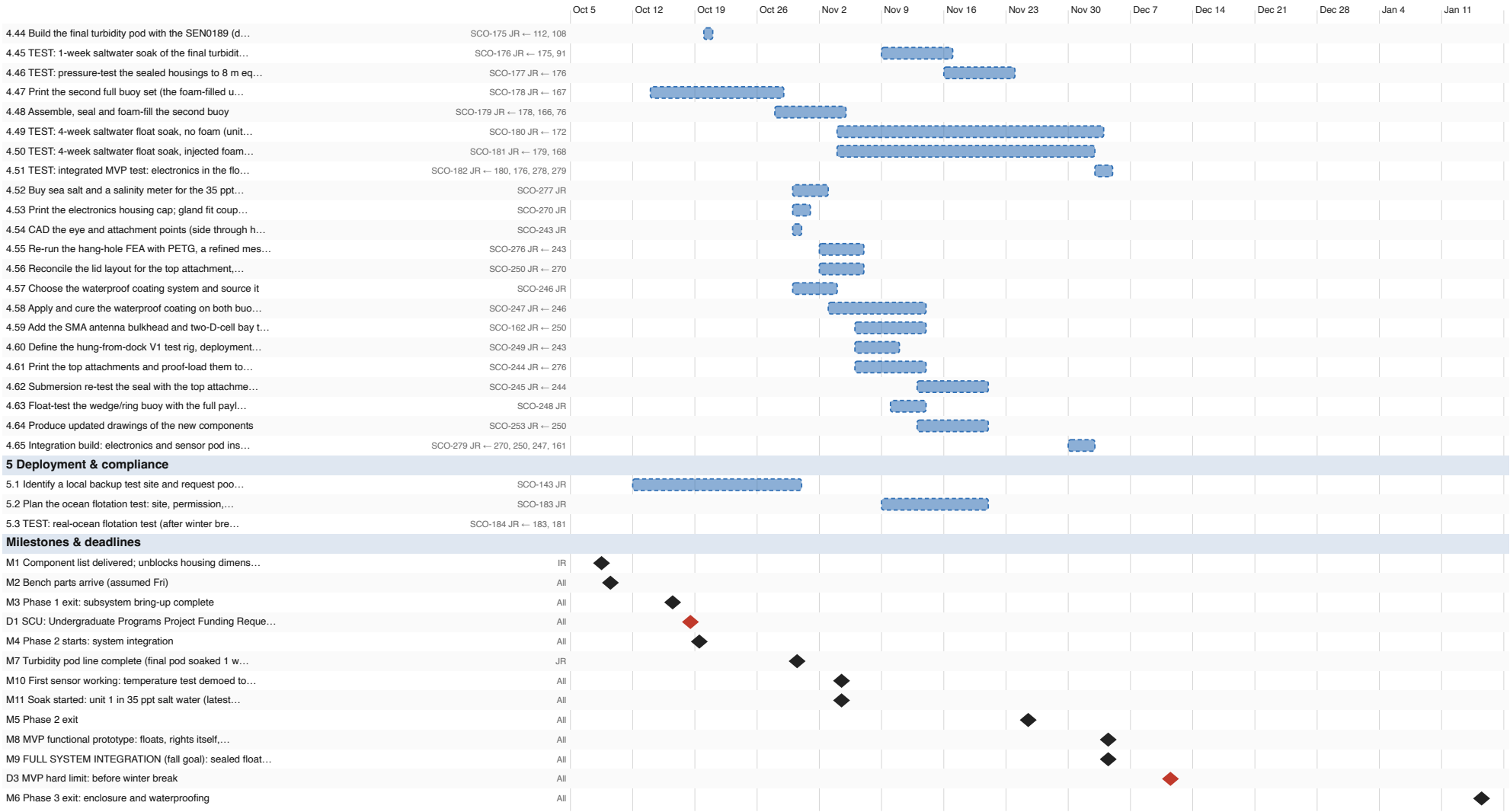


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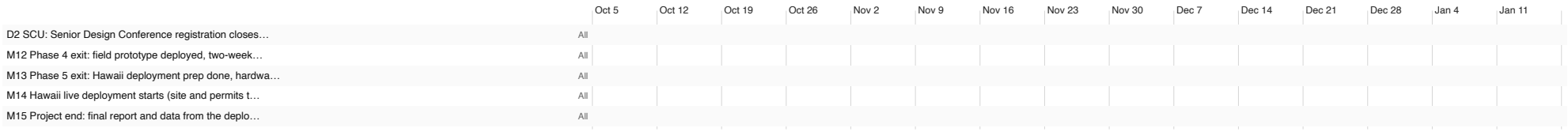


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Task table (all columns)

WBS	Linear	Task	Owner	Phase	Status	Start	End	Days	Depends on	Outside constraint	Milestone / deadline	Date source
1 Project management & documentation												
1.1	SCO-117	Submit the senior design budget (SCU funding request)	David	Phase 1	Todo	2026-10-06	2026-10-16	11		Lab budget was due the week of 2026-09-28 (overdue); SCU funding request deadline Oct 18; must match the parts order	Funding request submitted	Linear due
1.3	SCO-140	Map the layered test protocol and unit counts per stage	John	Phase 1	Todo	2026-10-10	2026-10-15	6		Advisor asked for a layered test plan (2026-09-30 meeting)	Test protocol drafted	Linear due
1.4		Gantt chart and WBS for the course assignment	John	Phase 1	In Progress	2026-10-06	2026-10-18	13		Course deadline Oct 18; whole team	Gantt submitted	derived
1.5	SCO-149	Write an alternatives analysis for the main mechanical choices	John	Phase 1	Backlog	2026-10-19	2026-11-06	19				estimated
1.6	SCO-90 / SCO-95	LLC and patent strategy for SCOUT IP	John	Phase 1	Backlog	2026-10-26	2026-11-25	31		Possible advisor or university IP office input		estimated
1.7	SCO-121	Decide a per-unit naming scheme for Scout prototypes	Unassigned	Phase 1	Backlog	2026-10-20	2026-10-30	11		Unassigned in Linear; needs an owner and a team decision		estimated
1.8	SCO-211	Get the hazard assessment form signed by Dr. Krishnan	John	Phase 2	Needs Decision	2026-10-28	2026-11-04	8		Form submitted 10/28; waiting on the advisor signature; blocks water-near-electronics lab work	Hazard form signed	estimated
1.9	SCO-260	Present the top-level system architecture and demos to Dr. Krishnan and Dr. Kuczenski	John	Phase 2	Todo	2026-11-02	2026-11-06	5		Fri 11/6, 6 PM proposed; all three members present	System-diagram review	Linear due
1.10	SCO-262	Apply for Environmental Engineering storage space for the water test tub	John	Phase 2	Todo	2026-11-02	2026-11-06	5		Needed before the 35 ppt float test and soak		Linear due
1.11	SCO-188	Write a black-box integration doc per subsystem	John	Phase 2	Todo	2026-11-02	2026-11-13	12		Each member writes their own; read by the integration build		estimated
2 Electrical hardware (ECE)												
2.1	SCO-70	Deliver the final electronics component list and dimensions	Isabella	Phase 1	Todo	2026-10-06	2026-10-08	3		Overdue since 2026-09-04; blocks housing dimensions	Component list final	Linear due
2.2	SCO-116	Take the full electronics design to Shoba for a power and solar review	Isabella	Phase 1	Todo	2026-10-08	2026-10-09	2		Shoba's availability	Design review complete	proposed
2.3	SCO-115	Bench parts order arrives (Feather M0, Adalogger, sensors, boost, TPL5111)	Isabella	Phase 1	Waiting on Parts	2026-10-09	2026-10-09	1	SCO-70, SCO-116	Vendor shipping; ordered through Dr. Krishnan 2026-10-05; arrival date assumed	Parts arrive	Linear due
2.4	SCO-142	Inventory spare sensors and boards in advisor and team labs	Isabella	Phase 1	Todo	2026-10-07	2026-10-12	6		Needs advisor and lab access		proposed
2.5	SCO-8	Decide the hydrophone part (H2a-XLR vs H2dM)	Isabella	Phase 1	Needs Decision	2026-10-12	2026-10-16	5		Hydrophone not in the bench order	Decision recorded	Linear due
2.6	SCO-10	Re-scope or close the LiFePO4 charging-path decision (ADR-0002)	Isabella	Phase 1	Needs Decision	2026-10-14	2026-10-19	6		Overtaken if ADR-0007 (Li-SOCl2 primary cell) is accepted; unblocks SCO-83	Decision recorded	proposed
2.7	SCO-150	Make it impossible to charge or short the primary D cell	Isabella	Phase 1	Todo	2026-10-09	2026-10-14	6	SCO-115	Fire-safety layers must exist before any real cell; bench uses a current-limited supply		Linear due
2.8	SCO-152	Bring up the bench breadboard: DS18B20, RTC, microSD, Grove turbidity wiring	Isabella	Phase 1	Todo	2026-10-12	2026-10-13	2	SCO-115	Needs the parts to arrive; bench supply and uA meter reserved	Bench bring-up complete	Linear due
2.9	SCO-23	Measure sleep current against the <5 mA target	Isabella	Phase 1	Waiting on Parts	2026-10-13	2026-10-14	2	SCO-152, SCO-115	uA-range multimeter; 3.6 V current-limited supply		Linear due
2.10	SCO-151	Prove TPL5111 power-gating cuts the sleep floor to ~20 uA	Isabella	Phase 1	Todo	2026-10-14	2026-10-15	2	SCO-23, SCO-115	Firmware cold-boot support from David (SCO-25)		Linear due
2.11	SCO-153	Measure the Grove turbidity sensor: current, settle time, inrush, polarity	Isabella	Phase 1	Todo	2026-10-14	2026-10-16	3	SCO-152, SCO-115	Oscilloscope for inrush		proposed
2.12	SCO-154	Measure LoRa transmit current at +11 dBm	Isabella	Phase 1	Todo	2026-10-16	2026-10-19	4	SCO-152	1 ohm shunt; re-check after SCO-98		proposed
2.13	SCO-155	Run the RFM95 link test with a second radio	Isabella	Phase 1	Todo	2026-10-15	2026-10-16	2	SCO-152, SCO-115	Second radio is not in the 10-05 order; lab stock or a new order	Radio link proven	Linear due

2.14	SCO-47	Design the SEN0189 analog front end as non-inverting	Isabella	Phase 1	In Progress	2026-09-01	2026-10-16	46	SCO-115	Polarity must not invert; bench check after parts arrive; Design started 2026-09-01; bench verification follows SCO-115 and SCO-153		Linear due
2.15	SCO-87	Implement per-sensor power switching via transistors	Isabella	Phase 1	Todo	2026-10-16	2026-10-21	6	SCO-152, SCO-153	V1-required on a primary cell		proposed
2.16	SCO-60	Add an internal temp/humidity sensor for State-of-Health	Isabella	Phase 1	Todo	2026-10-12	2026-10-15	4				proposed
2.17	SCO-156	Document the bench wiring: labeled photo and schematic sketch to /hardware	Isabella	Phase 1	Todo	2026-10-19	2026-10-20	2	SCO-152, SCO-153	Phase 1 exit item	Phase 1 hardware exit	proposed
2.18	SCO-83	Implement battery-voltage telemetry	Isabella	Phase 1	Blocked	2026-10-19	2026-10-23	5	SCO-10	Depends on the battery chemistry decision		Linear due
2.19	SCO-18 / SCO-84	Chlorophyll fluorometer feasibility and battery connector polarity check	Isabella	Phase 2	Deferred	2026-11-03	2026-11-25	23		Deferred in Linear; not this version		proposed
2.20		TEST: every sensor reads data (DS18B20 temperature and Grove turbidity) in a water bath, 24 hr log	Isabella	Phase 1	Proposed	2026-10-16	2026-10-22	7	SCO-152, SCO-153	Proposed by John, not in Linear; needs the bench parts and the SEN0189 front end	All sensors read	proposed
2.22		TEST: data transmission: a sensor packet goes over LoRa to the shore station and matches the logged value	Isabella	Phase 2	Proposed	2026-10-30	2026-11-06	8	SCO-155, SCO-24, SCO-98	Proposed by John, not in Linear; waits on the shore receiver (David, SCO-24, his dates unchanged)	Packet delivered end to end	proposed
2.23	SCO-158	Place the three follow-up electronics orders (D cells, hydrophone, outdoor antenna)	Isabella	Phase 2	Todo	2026-10-19	2026-10-26	8		Proposed dates; Linear due is Nov 6; needs the D cells in hand for SCO-159	Follow-up parts ordered	proposed
2.24	SCO-159	TEST: run 7 days on two D cells and confirm a 12-month extrapolation	Isabella	Phase 2	Todo	2026-11-09	2026-11-16	8	SCO-158	Proposed dates; Linear due is Nov 25; dependency proposed, not in Linear	Battery life extrapolated	estimated
2.25	SCO-161	TEST: run the sealed buoy electronics 7 to 14 days in the housing	Isabella	Phase 2	Todo	2026-11-16	2026-11-28	13	SCO-159	Proposed dates; Linear due is Dec 18; must end before Nov 30 for the MVP test; dependency proposed, not in Linear	Sealed run complete	estimated
2.26	SCO-258	Write the temperature sensor test protocol and send it to Shoba and Jes	Isabella	Phase 2	Todo	2026-10-28	2026-10-30	3		Due 10/29, late; becomes the template for every sensor	Protocol sent	Linear due
2.27	SCO-259	Run the temperature sensor test and demo it in the lab	Isabella	Phase 2	Todo	2026-10-30	2026-11-04	6	SCO-258	Wed 11/4 advisor demo; water on the table is fine	First sensor working	Linear due
2.28	SCO-189	Order the turbidity sensor and a spare	Isabella	Phase 2	Waiting on Parts	2026-10-26	2026-11-02	8		Mouser order in transit; blocks turbidity calibration, the pod soak and the MVP		estimated
2.29	SCO-216	Reuse the earlier team's hydrophone amplifier design	Isabella	Phase 2	Todo	2026-10-30	2026-11-06	8		Needs Ben's thesis design from Shoba and Dr. Wolf		Linear due
2.30	SCO-204	Calibrate the internal humidity sensor with known amounts of water	Isabella	Phase 2	Todo	2026-11-09	2026-11-13	5		Turns the 70 %RH placeholder into a measured threshold		Linear due
3 Firmware & software (CSEN)												
3.1	SCO-25	Validate the Phase-1 SAMD21 drivers on real hardware	David	Phase 1	Todo	2026-10-13	2026-10-16	4	SCO-115, SCO-152	Needs the Feather M0 from the bench order	Drivers validated on hardware	Linear due
3.2	SCO-24	Bring up the real LoRa receiver on the shore station	David	Phase 1	Done	2026-10-16	2026-10-30	15	SCO-115, SCO-155	Linear due 10-16 moved to 10-30: shore parts (Feather S3, LoRa module) are not ordered yet		Linear due
3.3	SCO-98	Bench-verify the FCC-compliant LoRa modem config	David	Phase 1	In Test	2026-10-16	2026-10-20	5	SCO-155, SCO-115	Config is merged; only bench verification remains		estimated
3.4	SCO-12	Calibrate the Grove turbidity sensor against formazin standards (FAU)	David	Phase 2	Blocked	2026-11-03	2026-11-20	18	SCO-189, SCO-153	Waits on the Grove sensor (SCO-189, in transit); Linear due 11/20		estimated
3.5	SCO-104	Faculty or a compliance lab review of the FCC 915 MHz findings	David	Phase 1	Todo	2026-10-12	2026-10-23	12		Reviewer availability		estimated
3.6	SCO-26 / SCO-27	Fix the run_pipeline flag inconsistency and recover the missing Sesoko recording	David	Phase 1	Backlog	2026-10-19	2026-10-30	12		Recording may need the original data source		estimated
3.7	SCO-119 / SCO-97	Adopt Navid's sleep/wake protocol and re-derive the packet ceiling	David	Phase 2	Backlog	2026-10-26	2026-11-13	19				estimated
3.8	SCO-120 / SCO-57 / SCO-58	Review the IoT dashboard, add biofouling state to the Fleet page, improve the PNG dashboard	David	Phase 2	Backlog	2026-11-02	2026-11-25	24			Phase 2 software exit	estimated

3.9	SCO-20 / SCO-67	Drift reference and acoustic tamper detection	David	Phase 2	Deferred	2026-11-09	2026-11-25	17		Deferred in Linear; not this version		estimated
3.10	SCO-160	TEST: run the standby duty-cycle loop unattended for 24 hr on the bench	David	Phase 2	Todo	2026-11-02	2026-11-06	5		Linear due date; start estimated; not rescheduled	Standby loop proven	Linear due
3.11	SCO-237	Push received readings from the shore station to the dashboard over WiFi	David	Phase 2	In Test	2026-10-28	2026-11-06	10		Merged 10/29; bench-unverified		estimated
3.12	SCO-278	TEST: end-to-end rehearsal, buoy reading to shore station to dashboard, 48 h on the bench	David	Phase 2	Backlog	2026-11-09	2026-11-20	12	SCO-160, SCO-237, SCO-152	Needs the bench bring-up; feeds the integrated MVP test	Full chain proven	Linear due
4 Mechanical & structure (GENG)												
4.1	SCO-105	Reprint the face-seal clamp and record the failing O-ring tolerance	John	Phase 1	Todo	2026-10-08	2026-10-12	5		Printer access		Linear due
4.3	SCO-108	Submersion re-test the face-seal joint on the clamp and sensor pod	John	Phase 1	In Test	2026-10-12	2026-10-16	5	SCO-105	24 hr soaks; RR-14 result 10-07; RR-15 (6 bolts) after; Coupon re-tests (RR-14, RR-15) run now; the clamp re-test follows SCO-105 and SCO-106	Face seal passes a 24 hr soak	estimated
4.5	SCO-112	Grow the sensor pod dry chamber to 52 mm and re-export the penetration and flood chamber	John	Phase 1	In Progress	2026-09-14	2026-10-13	30		Epoxy-potted lead penetration		Linear due
4.6	SCO-91	Validate the remodelled turbidity sensor housing face seal	John	Phase 1	Blocked	2026-10-20	2026-10-23	4	SCO-108			Linear due
4.7	SCO-141	Start saltwater-bath leak tests on printed test capsules	John	Phase 1	Todo	2026-10-08	2026-10-15	8		1 week soak at about 35 ppt		Linear due
4.10	SCO-107	Finish the electronics housing lid: bolt clearance holes and marking wording	John	Phase 1	Done	2026-10-13	2026-10-15	3				Linear due
4.11	SCO-103	Engrave the FCC ID label into the enclosure before printing	John	Phase 1	Done	2026-10-09	2026-10-12	4		Engraved before the final print (related to SCO-107 and SCO-93, not blocked by them)		Linear due
4.12	SCO-122	Order the stainless fasteners once the housing is dimensioned	John	Phase 1	Blocked	2026-10-14	2026-10-17	4		Vendor lead time; ordered only after SCO-49 fixes the housing dimensions		Linear due
4.13	SCO-68	Design the chassis top section: sealing and service-access port	John	Phase 1	In Progress	2026-09-03	2026-10-21	49	SCO-107, SCO-105			Linear due
4.17	SCO-63	Bring the large lab 3D printer back online	John	Phase 1	Backlog	2026-10-07	2026-10-16	10		Lab technician or advisor help		Linear due
4.18	SCO-76	Run foam-fill manufacturing trials on sacrificial wedges	John	Phase 1	Deferred	2026-10-19	2026-10-21	3	SCO-166	Printer access for sacrificial wedges		Linear due
4.20	SCO-124	Rebuild the freeboard model's vertical stack from the STEP geometry	John	Phase 1	Done	2026-09-29	2026-10-12	14		v7 STEP committed 2026-10-06		estimated
4.28	SCO-77	Define the marine fastener and anti-loosening strategy	John	Phase 1	Backlog	2026-10-16	2026-10-16	1				Linear due
4.29	SCO-72	Identify a propeller-resistant mooring/tether line	John	Phase 1	Backlog	2026-10-21	2026-10-30	10				estimated
4.30	SCO-94 / 144 / 145 / 146 / 147 / 148	Design improvement studies (inserts, cable routing, modular rod, split chassis, float-free, manufacturability)	John	Phase 1	Backlog	2026-10-27	2026-11-25	30				estimated
4.31	SCO-54 / SCO-56	Refine the stem and solar mount; audit the Onshape Initial Frame folder	John	Phase 1	Backlog	2026-10-19	2026-11-25	38			Phase 2 mechanical exit	Linear due
4.32	SCO-61 / SCO-66	Design the snap-connector sensor string; produce an exploded-view animation	John	Phase 2	Backlog	2026-11-03	2026-11-25	23				Linear due
4.33	SCO-92 / SCO-64 / SCO-78 / SCO-79	Corrosion strategy, material samples, marine coating, biofouling features	John	Phase 3	Todo / Backlog	2026-11-30	2027-01-15	47		Winter break; printer access	Phase 3 exit	estimated
4.35	SCO-166	Source the 2 lb pour foam (US Composites #0204, about 3 kits)	John	Phase 1	Deferred	2026-10-08	2026-10-16	9		No foam on hand; vendor shipping time; unblocks the foam trials and the foam-filled buoy	Foam on hand	Linear due
4.36	SCO-167	Source PETG filament for the second full buoy (about 4 to 5 kg, to confirm)	John	Phase 1	Waiting on Parts	2026-10-08	2026-10-14	7		Vendor shipping time; the second print waits on it	Filament on hand	Linear due
4.38	SCO-169	Finish J-B Weld on the remaining exterior seams of the assembled buoy	John	Phase 1	Todo	2026-10-12	2026-10-12	1		Epoxy cure time before water		Linear due

4.39	SCO-170	TEST: dry-float the buoy in a bin; waterline and freeboard vs the 4.27 in model	John	Phase 1	Done	2026-10-16	2026-10-16	1	SCO-124, SCO-169	Big plastic bin; benchmark within +-0.5 in (placeholder until SCO-139)	First float check	Linear due
4.40	SCO-171	TEST: self-righting tilt test from 90 and 180 degrees	John	Phase 2	Todo	2026-11-02	2026-11-03	2		Proposed 11/2-11/3 (Linear due 11/17 is too late for the soak); needs the printed buoy and a tub	Buoy rights itself	estimated
4.41	SCO-172	TEST: float the whole buoy in 35 ppt salt water; record the waterline shift	John	Phase 2	Todo	2026-11-03	2026-11-04	2	SCO-171, SCO-277	Proposed; needs salt, meter and the tub (SCO-277, SCO-262); salinity verified at 35 ppt first	Saltwater float passed	estimated
4.42	SCO-173	TEST: drop a sacrificial assembled ring with 1.5 in caps	John	Phase 2	Todo	2026-10-19	2026-10-22	4	SCO-170	Spare parts only, not the soak buoy		Linear due
4.44	SCO-175	Build the final turbidity pod with the SEN0189 (dry chamber 52 mm, 6 bolts)	John	Phase 1	Todo	2026-10-20	2026-10-20	1	SCO-112, SCO-108	SEN0189 arrives with the bench order; printer access	Final pod built	Linear due
4.45	SCO-176	TEST: 1-week saltwater soak of the final turbidity pod with the Grove sensor live	John	Phase 2	Todo	2026-11-09	2026-11-16	8	SCO-175, SCO-91	Needs the Grove sensor (SCO-189, waiting on parts); Linear due 11/24	Turbidity pod line complete	estimated
4.46	SCO-177	TEST: pressure-test the sealed housings to 8 m equivalent for 24 hr	John	Phase 2	Todo	2026-11-16	2026-11-23	8	SCO-176	Proposed; Linear due 11/9 cannot hold if it follows the pod soak		estimated
4.47	SCO-178	Print the second full buoy set (the foam-filled unit)	John	Phase 1	Deferred	2026-10-14	2026-10-28	15	SCO-167	Industrial printer or Prusa XL windows; chassis alone is about 35 hr	Second buoy printed	Linear due
4.48	SCO-179	Assemble, seal and foam-fill the second buoy	John	Phase 2	Deferred	2026-10-28	2026-11-04	8	SCO-178, SCO-166, SCO-76	Foam trial results; full cure before water	Foam buoy ready	Linear due
4.49	SCO-180	TEST: 4-week saltwater float soak, no foam (unit 1)	John	Phase 2	Todo	2026-11-04	2026-12-03	30	SCO-172	Proposed: must start by 11/5 to end 12/3; Linear due 12/11; 4 weeks, weekly weigh-in, Thanksgiving week	Unit 1 soak complete	estimated
4.50	SCO-181	TEST: 4-week saltwater float soak, injected foam (unit 2)	John	Phase 2	Deferred	2026-11-04	2026-12-02	29	SCO-179, SCO-168	Same bin routine as unit 1; must end by Dec 4	Foam vs no-foam comparison	Linear due
4.51	SCO-182	TEST: integrated MVP test: electronics in the floating buoy, sensors read, packet at the shore station	John	Phase 2	Todo	2026-12-03	2026-12-04	2	SCO-180, SCO-176, SCO-278, SCO-279	Proposed; waits on the soak, the pod soak, the end-to-end rehearsal and the integration build; hard limit Dec 11	MVP functional prototype	estimated
4.52	SCO-277	Buy sea salt and a salinity meter for the 35 ppt tests	John	Phase 2	Todo	2026-10-30	2026-11-02	4		Critical path: every day of delay moves the soak and the MVP		Linear due
4.53	SCO-270	Print the electronics housing cap; gland fit coupon first	John	Phase 2	In Progress	2026-10-30	2026-10-31	2		Printing 10/30; log weight and seal fit		Linear due
4.54	SCO-243	CAD the eye and attachment points (side through holes)	John	Phase 2	Todo	2026-10-30	2026-10-30	1		Finishing 10/30		Linear due
4.55	SCO-276	Re-run the hang-hole FEA with PETG, a refined mesh and the right loads	John	Phase 2	Todo	2026-11-02	2026-11-06	5	SCO-243	Pass mark SF >= 4 at the hole edge	Hang point validated	estimated
4.56	SCO-250	Reconcile the lid layout for the top attachment, SMA bulkhead and two-D-cell bay	John	Phase 2	Todo	2026-11-02	2026-11-06	5	SCO-270	Uses the cap STEP with sideways glands		Linear due
4.57	SCO-246	Choose the waterproof coating system and source it	John	Phase 2	Todo	2026-10-30	2026-11-03	5		Choose 10/30; buy tentative coating this weekend		Linear due
4.58	SCO-247	Apply and cure the waterproof coating on both buoys	John	Phase 2	Todo	2026-11-03	2026-11-13	11	SCO-246	Cure time per the coating's data sheet		Linear due
4.59	SCO-162	Add the SMA antenna bulkhead and two-D-cell bay to the lid	John	Phase 2	Todo	2026-11-06	2026-11-13	8	SCO-250			Linear due
4.60	SCO-249	Define the hung-from-dock V1 test rig, deployment and retrieval	John	Phase 2	Todo	2026-11-06	2026-11-10	5	SCO-243			Linear due
4.61	SCO-244	Print the top attachments and proof-load them to about 1.5x the hung load	John	Phase 2	Todo	2026-11-06	2026-11-13	8	SCO-276			Linear due
4.62	SCO-245	Submersion re-test the seal with the top attachment installed	John	Phase 2	Todo	2026-11-13	2026-11-20	8	SCO-244			Linear due
4.63	SCO-248	Float-test the wedge/ring buoy with the full payload	John	Phase 2	Todo	2026-11-10	2026-11-13	4				Linear due
4.64	SCO-253	Produce updated drawings of the new components	John	Phase 2	Todo	2026-11-13	2026-11-20	8	SCO-250			Linear due

4.65	SCO-279	Integration build: electronics and sensor pod installed in the printed buoy and sealed	John	Phase 2	Backlog	2026-11-30	2026-12-02	3	SCO-270, SCO-250, SCO-247, SCO-161	Proposed; needs the sealed electronics run to finish 11/28	Complete buoy assembled	estimated
5 Deployment & compliance												
5.1	SCO-143	Identify a local backup test site and request pool access	John	Phase 1	Backlog	2026-10-12	2026-10-30	19		Site owner's permission		estimated
5.2	SCO-183	Plan the ocean flotation test: site, permission, boat or pier, safety	John	Phase 2	Todo	2026-11-09	2026-11-20	12		Site owner or harbour permission; the test itself is after winter break	Ocean test planned	Linear due
5.3	SCO-184	TEST: real-ocean flotation test (after winter break)	John	Phase 4	Todo	2027-01-25	2027-01-29	5	SCO-183, SCO-181	Winter break; sea state and permit window	Ocean flotation passed	Linear due
Milestones & deadlines												
M1		Component list delivered; unblocks housing dimensions	Isabella	Phase 1		2026-10-08	2026-10-08	1			Component list final	estimated
M2		Bench parts arrive (assumed Fri)	All	Phase 1		2026-10-09	2026-10-09	1		Assumption; confirm with Isabella	Parts arrive	Linear
M3		Phase 1 exit: subsystem bring-up complete	All	Phase 1		2026-10-16	2026-10-16	1		Phase 1 project window	Phase 1 exit	estimated
D1		SCU: Undergraduate Programs Project Funding Request due; Gantt chart due	All	Phase 1		2026-10-18	2026-10-18	1		SCU Senior Design Resources page	Funding request and Gantt submitted	Linear
M4		Phase 2 starts: system integration	All	Phase 2		2026-10-19	2026-10-19	1				estimated
M7		Turbidity pod line complete (final pod soaked 1 week)	John	Phase 2		2026-10-30	2026-10-30	1		Gate: SCO-176 passes	Pod line complete	estimated
M10		First sensor working: temperature test demoed to the advisors	All	Phase 2		2026-11-04	2026-11-04	1			First sensor working	Linear
M11		Soak started: unit 1 in 35 ppt salt water (latest start to finish by 12/3)	All	Phase 2		2026-11-04	2026-11-04	1			Soak started	estimated
M5		Phase 2 exit	All	Phase 2		2026-11-25	2026-11-25	1			Phase 2 exit	estimated
M8		MVP functional prototype: floats, rights itself, reads one temperature and one turbidity value, packet at the shore station (target)	All	Phase 3		2026-12-04	2026-12-04	1		Decided 2026-10-07	MVP target	estimated
M9		FULL SYSTEM INTEGRATION (fall goal): sealed floating buoy reads temperature and turbidity, packet reaches the shore station and the dashboard	All	Phase 2		2026-12-04	2026-12-04	1		Linear milestone: Full system integration (fall goal); hard limit Dec 11	Full system integration	Linear
D3		MVP hard limit: before winter break	All	Phase 3		2026-12-11	2026-12-11	1		Decided 2026-10-07; Dec 4 is the target		estimated
M6		Phase 3 exit: enclosure and waterproofing	All	Phase 3		2027-01-15	2027-01-15	1			Phase 3 exit	estimated
D2		SCU: Senior Design Conference registration closes (winter)	All	Phase 4		2027-02-01	2027-02-01	1		SCU Senior Design Resources page		Linear
M12		Phase 4 exit: field prototype deployed, two-week local test done	All	Phase 4		2027-02-26	2027-02-26	1				Linear
M13		Phase 5 exit: Hawaii deployment prep done, hardware shipped	All	Phase 5		2027-03-19	2027-03-19	1				Linear
M14		Hawaii live deployment starts (site and permits to be confirmed)	All	Phase 6		2027-03-22	2027-03-22	1				Linear
M15		Project end: final report and data from the deployment	All	Phase 6		2027-05-28	2027-05-28	1				Linear

Date source: **Linear due** = the end date equals the issue's Linear due date; **estimated** = proposed from the issue text, parts timing or the phase window and awaiting the owner's confirmation; **derived** = set by this assignment; **proposed** = a new date John proposes for the task's owner, shown only until the owner agrees and Linear is updated. Days are calendar days, inclusive.